# Brian Fung

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## Summary

DevOps / Platform Engineer with 9 years across technology, including platform engineering, CI/CD and test automation. I build infrastructure on AWS and Google Cloud Platform with Terraform and AWS CDK, and edge services on Akamai EdgeWorkers. Focused on cutting costs, improving reliability and building self-service tools that remove engineering bottlenecks. Currently learning Kubernetes and Microsoft Azure, and experimenting with how prompt loops can accelerate development workflows.

## Professional Experience

**Kmart Australia**
*DevOps Engineer* July 2024 – Present

**Cost optimisation and FinOps**

- Initiated and delivered a Redis to Valkey migration on AWS ElastiCache ahead of Redis end-of-life, avoiding extended support charges and reducing caching costs by 70% (approximately AUD 20,000 per year from January 2027), with no application rewrites.
- Implemented Amazon ECR lifecycle policies to automatically expire unused container images, reducing storage costs by approximately AUD 10,000 per year.
- Contributed to the Imperva to Akamai CDN and WAF harmonisation for kmart.com.au, consolidating vendors and reducing costs across the Kmart Group.

**Platform and AI infrastructure**

- Delivered Kmart's first production applications on Google Cloud Platform, Joy (a Gemini-powered generative AI shopping assistant) and Vertex AI Search for Commerce, taking each from proof of concept to production in 3 weeks on aggressive timelines, using newly released GCP AI products with many unknowns.
- Built end-to-end GCP infrastructure, CI/CD pipelines and CDN configuration for Joy using Terraform.
- Built GCP infrastructure for the Constructor.io to Vertex AI Search for Commerce migration, including Cloud Run, load balancers, Cloud Armor, Certificate Manager, a token service, canary deployments and Akamai traffic routing.
- Negotiated GCP security architecture with the security team (DNS, internal domain reuse, Google-managed certificate ownership) and enabled keyless Buildkite deployments via Workload Identity Federation (WIF).
- Diagnosed and remediated performance issues, including approximately 1,900 ms of search latency caused by CDN tiered distribution routing through US points of presence, and drove the APAC fix with Akamai.
- Led a VPC migration of the consent management system, integrated with Tealium and Braze, using a blue/green deployment strategy, the first use of blue/green deployment within Kmart Digital, achieving zero downtime.

**Edge engineering and security**

- Led solution architecture for Kasada bot protection using Akamai EdgeWorkers Flexible Composition, with automated regression coverage and fault injection tests to validate resilience and failure handling.
- Negotiated with the security team to enable Akamai pragma debug headers under IP allowlisting and required-header controls, unblocking automated test coverage that guards CDN and EdgeWorker functionality from regressions.
- Main contributor to a CDN-as-code repository, managing Akamai property configuration and EdgeWorkers as code, with automated regression tests in CI/CD.
- Delivered Kmart's first production Akamai EdgeWorker (visitor ID), and led the investigation that showed true median latency was about 95 ms rather than the reported 800 ms, correcting a measurement error in RUM instrumentation.
- Supported Akamai WAF and Bot Manager tuning and a firewall cutover.

**Developer enablement and self-service**

- Identified the need for and built a self-service Akamai cache purge tool for the online merchandising team, secured with Microsoft Entra ID single sign-on (SSO) and a full audit trail, cutting purge turnaround from 10–15 minutes (or next business day after hours) to seconds. The team ran 695 purges in the first 3 months, saving an estimated 115–170 hours of waiting and 695 engineer interruptions.
- Built a self-service Redis cache purge tool so teams can clear cached data without engineering involvement.
- Built a Playwright test automation framework integrated with CI/CD and adopted by 4 squads, migrating from WebdriverIO and cutting regression suite runtime by 75%, from 40 minutes to 10 minutes.
- Built a serverless personalisation API platform on AWS with GitHub Actions, CDK scaffolding for API Gateway and Lambda endpoints, and Sumo Logic observability built in.
- Built an Express mock service on Amazon ECS so teams can develop against endpoints before real services exist.

**Performance and reliability**

- Led Availability Zone (AZ) disaster recovery testing for the Kmart website using AWS Fault Injection Service, covering scoping, observability dashboards, performance tests and automated functional tests to validate recovery.

- Identified and remediated the platform issues that had prevented recovery from an AZ failure, taking the platform from no recovery to full recovery within a 3-minute Recovery Time Objective (RTO).
- Led performance testing for Peak trading readiness using JMeter, validating platform capacity ahead of the highest-traffic period of the year.
- Designed a token toggle supporting both real and forged authentication tokens, with production safeguards, enabling load testing of authenticated customer journeys without disrupting test environments.
- Built test data generation scripts for performance testing, reducing test preparation time from hours to minutes.
- Participated in incident response as part of the on-call rotation, and led the root cause analysis (RCA) of a cascading failure across production ECS microservices.
- Led the Recommender API migration for service stability, and built New Relic and Sumo Logic observability.

**Leadership and mentoring**
- Mentored a graduate engineer and built a dotfiles bootstrapper to speed up engineer onboarding.

**Kmart Australia**
*Software Test Engineer* September 2021 – July 2024

- Core maintainer of the Kmart Online test automation framework, migrating WebdriverIO from synchronous to asynchronous and adding linting, logging and end-to-end email testing, improving suite stability.
- Moved the automation suite from manual QA-triggered runs into Buildkite CI/CD, enabling automated regression testing on every pipeline run.
- Migrated functional tests to run against stubs using WireMock, reducing runtime by 90%, from 30 minutes to 3 minutes for 50 tests.
- Core contributor to the performance test suite for Peak trading readiness.

**Two Bulls**
*Senior QA Engineer* September 2019 – August 2021
*QA Engineer* May 2017 – September 2019

- Created and maintained CI/CD pipelines for iOS, Android, web and game engine projects using Jenkins, GitLab CI/CD, Bitrise, GitHub Actions and Fastlane, including code signing and automated release deployment to the App Store and Google Play.
- Migrated Jenkins jobs to version-controlled declarative pipelines (pipeline as code), and maintained the on-premises Jenkins build infrastructure for a cross-platform video SDK.
- Designed an automated smoke testing solution on AWS (Lambda, S3, CloudWatch, CodePipeline, Device Farm), provisioned with CloudFormation.
- Built scripts to automate test build installation on iOS, Android and embedded devices, cutting test setup time from minutes to seconds.
- Owned test strategy, test creation, execution and automation for web and native mobile applications in agile and waterfall environments, using Cypress, Appium and XCUITest.
- Selected clients: Service Victoria, Beyond Blue, Scouts Australia, Verizon, Sesame Street, The Wiggles, 28 By Sam Wood.

## Technical Skills

**AWS:** EC2, ECS, ECR, Lambda, API Gateway, Application Load Balancer, VPC, Route 53, S3, CloudFront, IAM, DynamoDB, ElastiCache, SQS, SNS, Systems Manager, Fault Injection Service
**Google Cloud Platform:** Cloud Run, Cloud Load Balancing, Cloud Armor, VPC, IAM, Workload Identity Federation, Cloud Storage, Storage Transfer Service, BigQuery, Cloud Monitoring, Vertex AI Search, Gemini Enterprise for Customer Experience
**Infrastructure as Code:** Terraform, AWS CDK, CloudFormation
**CDN:** Akamai EdgeWorkers, Flexible Composition, Property Manager, WAF, Bot Manager; Imperva
**CI/CD:** Buildkite, GitHub Actions, GitLab CI/CD, Bitrise, Jenkins, AWS CodePipeline, Bitbucket Pipelines, Fastlane; blue/green and canary deployments
**Languages and platforms:** TypeScript, Python, Bash, Git, Linux, Docker
**Testing:** Playwright, Jest, Vitest, Supertest, Cypress, Appium, Cucumber, WireMock
**Performance testing:** JMeter, k6, Locust, Lighthouse
**Observability:** New Relic, Datadog, Sumo Logic, Amazon CloudWatch
**Security and identity:** AWS Secrets Manager, Wiz, Snyk, Kasada, Auth0, Microsoft Entra ID, SSO
**APIs, data and integrations:** REST, GraphQL, Redis, Valkey, Contentful, Tealium, Braze
**AI and developer tools:** Cursor CLI (AI coding agent), AWS DevOps Agent, AWS Security Agent, Neovim, tmux
**Practices:** TDD, BDD, pair programming, code review, Agile (Scrum, Kanban), CI/CD, infrastructure as code, incident response and on-call, root cause analysis, disaster recovery testing, FinOps

## Education and Training

**RMIT University** – Bachelor of Electrical Engineering / Bachelor of Commerce 2012 – 2017
**OWASP AppSec Day** – Web Hacking Essentials November 2019